Reproducibility map for "What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs"

Source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source	Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`	recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`	recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`	recomputed
Figure S7	`make_figs2.py`	recomputed
$\delta_{\min}(3)$, Table S11 recomputed columns, 61.14, 29.94	`recompute_theory.py` → `R_P1_theory_constants.json`	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures S6–S1	`make_figs3.py` → `R_P1_pulse_benchmarks.json`	recomputed
Table S28, Table S30, worked example, passive simulation	`recompute_extra.py` → `R_P1_extra.json`	recomputed
Daily noise (Table S14)	`P1.3-T1_sigma_obs`	real data
Table S29 recomputed columns	same	recomputed
K-cohort slopes (Table S31)	`T1_K_cohort_rate`	recomputed
ESPRIT slope −6.15	`P1.2-T1_esprit_rate`	recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`	recomputed
Passive slopes −4.21, −5.93	`T3_passive_rate`	recomputed
Tagged-flow slope 0.011	`T4_disclosure`	recomputed
Spectral blindness gains	`T5_spectral_blindness`	recomputed
Noise and capital grid (Table S13)	`P1.3-T1_sigma_obs`	real data

Result	Source		Kind
Loop gain and placebo (Table S19)	`P1.6-T1_ema_placebo`		real data
Estimator table (Table S20)	`P1.5-T1`, `T2`, `T3`		recomputed
Data integrity (111 files)	`R2-T9_data_integrity`		audit
Tables S10, S11 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` `R_P1_capital_ledger.json`	→	recomputed
Table 2; resolution floors; Table S2	`recompute_p1_protocols.py` `R_P1_protocols.json`	→	recomputed
SPRT capital figures	`recompute_p1_sprt.py` `R_P1_sprt.json`	→	recomputed
Table S8 (N^*, σ_{crit})	`recompute_crb.py` `R_P1_crb.json`	→	recomputed
Table S8 (ML columns), Table S29 (estimators)	`recompute_p1_estimators.py`		recomputed
Symmetric multi-asset network (Supplement S12)	`recompute_oct20.py`		recomputed
Proof audit: K_* bound and d_2 identity (Theorem 4.1), moment gaps (Theorem 5.1), spectral-blindness example (Theorem S12.1)	`recompute_audit.py` `R_audit_oct2026.json`	→	recomputed
Theorem 5.4, Table S32	`kcohort_upper.py` `R_P1_kcohort_upper.json`	→	recomputed
Lem. S17.1, Thm S17.2, Tab. S23	`general_minimax.py`		recomputed
Theorem 6.2 numbers	`disclosure_value.py` `R_P1_disclosure_value.json`	→	recomputed
Tables S17, S18, S21	Part 8: `V2_spot_panel`, `V3_fee_events`		real data
Table S25, top panel	Part 4: `X9_disclosure_metrics`		real data
Table S25, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…)		real data
Table S24	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…)		real data

Result	Source	Kind
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data
Theorem S9.4 full bound (Table S12), direct minimization	`hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json`	recomputed
Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant	`hardening/p1_checks.py` → `R_P1_hardening_checks.json`	recomputed
Stylized nonlinear and trained online rules (Table S15, Figure S4)	`recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…)	recomputed
Disclosure in dollars (Table S9)	`recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json`	recomputed
Out-of-sample post-shock return test (Supplement S19)	`recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…)	real data
Out-of-sample cohort-state trajectory test (Table S26)	`recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…)	real data
Proposition 6.4, Table S33	`theory/targeted_intervention.py` → `R_P1_targeted_intervention.json`	recomputed
Table S29, column N^*_{CRB}; Table S8, spacing-0.10 three-cohort row	`breakthrough_experiments/results_exp17_minimax_lecam_bou` `results_exp15_cohort_information.json`	recomputed (canonical run)

Result	Source	Kind
Table S12, best feasible probes and $L = 320, 480$	`R_P1_capital_ledger.json`, `R_P1_cquadL.json` (reproduced by `hardening/rh_quad.py`)	recomputed
Table S10, peak margins of the eigenvector probes	`breakthrough_experiments/results_exp14_capital_frontier.json`	recomputed (earlier run)
Table S23	`theory/R_P1_general_minimax.json`	recomputed
Table S29 and Table S8, estimator columns	`R_P1_estimators.json`, `R_P1_estimators_crb.json`	recomputed
Theorem 3.1, numerical check of the equivalence	`R2-T5_merged_invariance`	recomputed
Supplement S19 and Table 1: horizon-matched test on 30 new markets (29 of 29, median 7.3, smallest 2.6) and jackknife-corrected persistence (1.0008)	`thesis/evidence/C2_results.json` (specification `thesis/checks/prereg/C2_margin_vs_identify.json`)	real data
Supplement S19: exploratory 48- and 96-lag refit (median 7.5)	`thesis/evidence/pilot3_horizon_matched_results.json`	real data (exploratory)
Supplement S19 and Table 1: 105-day follow-up (median ratio 74.9; 9 identified; 10 of 30, 23 required)	`thesis/evidence/E1v2_results.json`	real data
Supplement S19 and Table 1: 58-day follow-up (22 of 29; 20 of 29, 22 required)	`thesis/evidence/D1_results.json`	real data
Supplement S16, Tables S22, S3 and 1: real-noise injection (slopes -6.01, -9.81; tagged 256–724)	`p1_injection/injection_test.py`, `p1_injection/results/I1_injection_test.json` (specification `p1_injection/prereg/P1_injection.json`, 596a700d…)	semi-synthetic
Supplement S3.7 and Table S4: numerical stress tests	`p1_stress/stress_theorems.py`, `p1_stress/results/S1_stress_tests.json`	numerical check

Result	Source	Kind
Theorem 3.1(i), Theorem 5.3 and Proposition 6.1 (machine-checked parts)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedModel.lean`, `Paper1Identification.lean`, `Paper1MomentGap.lean`, `Paper1PoleCount.lean`, `Paper1FiniteIdentification.lean`, `Paper1Tagged.lean`, `Paper1Testing.lean`, `Paper1LowerBound.lean`, `Paper1MatchedPair.lean`	Lean 4 proof
Theorem 5.4 (machine-checked, every K)	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PronyUpper.lean`, `Paper1PronyMoment.lean`, `Paper1PronyDeconv.lean`, `Paper1UniformInverse.lean`	Lean 4 proof
Theorems 3.1(ii), S12.1, 5.5 (zeros), 6.2(ii) (tagged upper bound) and Proposition 6.1 (instrument determinant), machine-checked parts	`research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedSeparation.lean`, `Paper1Blind.lean`, `Paper1MinPhase.lean`, `Paper1TaggedRates.lean`, `Paper1TaggedIV.lean`	Lean 4 proof
Supplement S18: post hoc robustness (futures prices, roll weeks, 26- and 52-week blocks)	`p11/robustness_cot.py`, `p11res/results/C4R_cot_robustness.json`	real data (post hoc)

Result	Source	Kind
Supplement S18 and Table 1: two-root test (2 of 8; smaller root -0.16 to 0.21)	`p11/two_root_test.py`, `p11res/results/C6M_cot_two_root.json` (specification `p11/prereg_memory/P11M_two_root.json`, e6633588…)	real data
Supplement S18 and Table 1: class-flow test (3 of 8; $F \geq 10$ in 7; $w \neq 0$ in 6)	`p11/tagged_flow_test.py`, `p11res/results/C5F_cot_tagged_flow.json` (specification `p11/prereg_flow/P11F_cot_tagged_flow.json`, 052a63f2…)	real data
Finite-horizon decision appendix: primary decisions, cost–horizon sensitivities (819 rows), hard-pair requirements, assumption failures, cash-flow audit	`studies/finite_horizon_costs_20261029/run.py` → `results/final_full_20261030/`	real data
Finite-horizon appendix: independent checks and LP cross-check	`studies/finite_horizon_costs_20261029/validation/`	real data